

Empirical Methods in Finance

Section 10 - Gaussian and Non-Gaussian GARCH Models

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Objectives of the lecture

This section has six goals:

1. understand why ARCH models are often too short-memory for financial returns,
2. introduce Gaussian GARCH models and their main properties,
3. derive volatility forecasts from a GARCH(1,1),
4. discuss estimation, inference, diagnostics and asymmetric effects,
5. move from Gaussian to non-Gaussian conditional distributions,
6. evaluate volatility forecasts and implement a simple S&P 500 application.

The ARCH idea was:

$$r_t = \mu_t + \varepsilon_t, \quad (1)$$

$$\varepsilon_t = \sigma_t Z_t, \quad (2)$$

$$\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \cdots + \alpha_p \varepsilon_{t-p}^2. \quad (3)$$

The empirical issue is that volatility is highly persistent.

A large ARCH order may fit the data, but it is not parsimonious:

- too many lag coefficients,
- difficult interpretation,
- unstable forecasts,
- weak out-of-sample performance.

A useful way to think about volatility

Financial returns often have little linear predictability:

$$\text{Cov}(r_t, r_{t-k}) \approx 0. \quad (4)$$

But their squared returns are persistent:

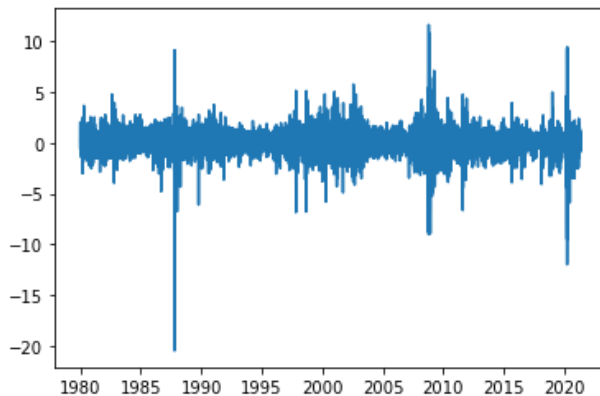
$$\text{Cov}(r_t^2, r_{t-k}^2) > 0. \quad (5)$$

The key object is therefore not only the expected return, but the full conditional distribution:

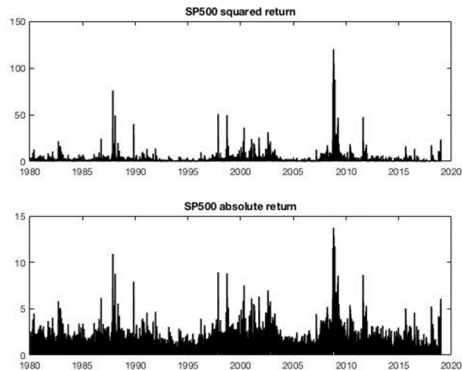
$$r_t \mid I_{t-1} \sim D(\mu_t, \sigma_t^2, \eta), \quad (6)$$

where η controls the shape of the innovation distribution.

S&P 500 returns: the volatility clustering problem



Squared returns: the object that GARCH tries to explain



Gaussian GARCH models

The GARCH model

Bollerslev's GARCH model adds lagged conditional variances to ARCH:

$$\varepsilon_t = \sigma_t z_t, \quad (7)$$

$$\sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^q \beta_j \sigma_{t-j}^2, \quad (8)$$

$$z_t \sim iid(0, 1). \quad (9)$$

The interpretation is simple:

- α_i measures the reaction to recent shocks,
- β_j measures the persistence of previous volatility,
- ω anchors the long-run variance.

Parameter restrictions

To keep the conditional variance positive, a standard sufficient condition is:

$$\omega > 0, \quad (10)$$

$$\alpha_i \geq 0, \quad i = 1, \dots, p, \quad (11)$$

$$\beta_j \geq 0, \quad j = 1, \dots, q. \quad (12)$$

For covariance stationarity:

$$\sum_{i=1}^p \alpha_i + \sum_{j=1}^q \beta_j < 1. \quad (13)$$

Then the unconditional variance is:

$$\text{Var}(\varepsilon_t) = \frac{\omega}{1 - \sum_{i=1}^p \alpha_i - \sum_{j=1}^q \beta_j}. \quad (14)$$

The GARCH(1,1) workhorse

The GARCH(1,1) is the most widely used specification:

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2. \quad (15)$$

Its long-run variance is:

$$\bar{\sigma}^2 = \frac{\omega}{1 - \alpha - \beta}. \quad (16)$$

The parameter $\alpha + \beta$ is the persistence of volatility:

- high α means strong reaction to new shocks,
- high β means slow decay,
- high $\alpha + \beta$ means long volatility cycles.

GARCH(1,1) as an ARMA model for squared returns

Let:

$$v_t = \varepsilon_t^2 - \sigma_t^2, \quad (17)$$

so that $\mathbb{E}[v_t \mid \mathcal{I}_{t-1}] = 0$.

Then:

$$\varepsilon_t^2 = \sigma_t^2 + v_t \quad (18)$$

$$= \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2 + v_t \quad (19)$$

$$= \omega + (\alpha + \beta) \varepsilon_{t-1}^2 + v_t - \beta v_{t-1}. \quad (20)$$

So squared returns follow an ARMA(1,1)-type representation.

The ARMA representation helps interpret GARCH:

$$\varepsilon_t^2 = \omega + \gamma \varepsilon_{t-1}^2 + v_t - \beta v_{t-1}, \quad \gamma = \alpha + \beta. \quad (21)$$

The GARCH(1,1) therefore behaves like a persistent model for variance shocks.

This explains why it can summarize volatility clustering with only three variance parameters:

$$\theta = (\omega, \alpha, \beta)'. \quad (22)$$

For a GARCH(1,1), the persistence of volatility shocks is:

$$\rho = \alpha + \beta. \quad (23)$$

The half-life of a volatility shock is approximately:

$$h = \frac{\log(0.5)}{\log(\rho)}. \quad (24)$$

Example:

- if $\rho = 0.95$, half-life is around 14 trading days,
- if $\rho = 0.99$, half-life is around 69 trading days.

Small differences in persistence have large effects on risk forecasts.

When:

$$\alpha + \beta \approx 1, \quad (25)$$

volatility shocks are extremely persistent.

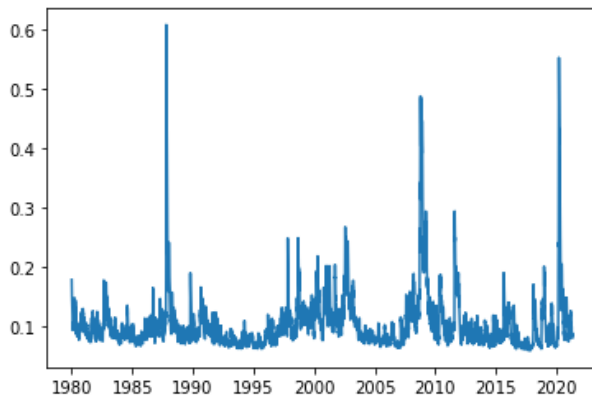
The IGARCH case imposes:

$$\alpha + \beta = 1. \quad (26)$$

Then the conditional variance has no finite unconditional variance in the usual sense, and shocks do not mean-revert in the standard GARCH way.

In practice, near-IGARCH estimates can also indicate omitted regime changes or structural breaks.

GARCH versus rolling and EWMA volatility



Forecasting and estimation

At date t , the one-step-ahead variance forecast is:

$$\mathbb{E}_t[\sigma_{t+1}^2] = \hat{\omega} + \hat{\alpha}\hat{\varepsilon}_t^2 + \hat{\beta}\hat{\sigma}_t^2. \quad (27)$$

This uses:

- the latest shock, $\hat{\varepsilon}_t^2$,
- the current conditional variance, $\hat{\sigma}_t^2$,
- the estimated reaction and persistence parameters.

The forecast is updated every day as new returns arrive.

For $h > 1$, the GARCH(1,1) forecast is recursive:

$$\mathbb{E}_t[\sigma_{t+h}^2] = \hat{\omega} + (\hat{\alpha} + \hat{\beta})\mathbb{E}_t[\sigma_{t+h-1}^2]. \quad (28)$$

Iterating forward:

$$\mathbb{E}_t[\sigma_{t+h}^2] = \bar{\sigma}^2 + (\hat{\alpha} + \hat{\beta})^{h-1} (\mathbb{E}_t[\sigma_{t+1}^2] - \bar{\sigma}^2). \quad (29)$$

Forecasts converge to the long-run variance when $\hat{\alpha} + \hat{\beta} < 1$.

Gaussian maximum likelihood

Assume:

$$z_t \sim N(0, 1). \quad (30)$$

Then:

$$\varepsilon_t \mid I_{t-1} \sim N(0, \sigma_t^2). \quad (31)$$

The conditional log-likelihood is:

$$\log L(\theta) = -\frac{1}{2} \sum_{t=1}^T \left[\log(2\pi) + \log(\sigma_t^2) + \frac{\varepsilon_t^2}{\sigma_t^2} \right]. \quad (32)$$

The variance path σ_t^2 is computed recursively for each candidate parameter vector.

A practical Gaussian GARCH estimation routine is:

1. choose starting values for $\mu, \omega, \alpha, \beta$,
2. initialize σ_0^2 with the sample variance,
3. recursively compute σ_t^2 ,
4. compute the conditional log-likelihood,
5. maximize it subject to parameter restrictions,
6. filter standardized residuals:

$$\hat{z}_t = \frac{\hat{\varepsilon}_t}{\hat{\sigma}_t}. \quad (33)$$

Under correct specification, the maximum likelihood estimator satisfies:

$$\sqrt{T}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, \mathcal{I}^{-1}), \quad (34)$$

where $\mathcal{I}$ is the information matrix.

In practice, the usual covariance estimator is based on the inverse Hessian:

$$\widehat{\text{Var}}(\hat{\theta}) = \left[-\frac{\partial^2 \log L(\hat{\theta})}{\partial \theta \partial \theta'} \right]^{-1}. \quad (35)$$

If the conditional distribution is misspecified, robust covariance estimators are preferable.

Testing for remaining ARCH effects

After estimating a mean model, run:

$$\hat{\varepsilon}_t^2 = a_0 + a_1 \hat{\varepsilon}_{t-1}^2 + \cdots + a_p \hat{\varepsilon}_{t-p}^2 + u_t. \quad (36)$$

The null hypothesis is:

$$H_0 : a_1 = \cdots = a_p = 0. \quad (37)$$

After estimating a GARCH model, the same test can be applied to standardized residuals:

$$\hat{z}_t^2. \quad (38)$$

A good volatility model should remove most serial dependence in squared standardized residuals.

For a fitted GARCH model, check:

- autocorrelation of standardized residuals $\hat{z}_t$,
- autocorrelation of squared standardized residuals $\hat{z}_t^2$,
- tail behavior of $\hat{z}_t$,
- stability of parameters through time,
- out-of-sample forecast performance.

A model can look good in-sample and still forecast poorly.

This is why volatility forecast evaluation is not a side topic.

Extensions of Gaussian GARCH

Why extend the basic GARCH model?

The basic Gaussian GARCH(1,1) captures volatility clustering, but it misses three common features:

1. **leverage effects**: negative returns often increase volatility more than positive returns,
2. **fat tails**: standardized residuals are rarely Gaussian,
3. **density asymmetry**: left-tail risk is often more important than right-tail risk.

The first issue is usually addressed through asymmetric GARCH models.

The second and third issues require non-Gaussian conditional distributions.

Empirically, bad news often has a larger effect on volatility than good news.

A simple news impact idea is:

$$\sigma_t^2 = f(\varepsilon_{t-1}, \sigma_{t-1}^2). \quad (39)$$

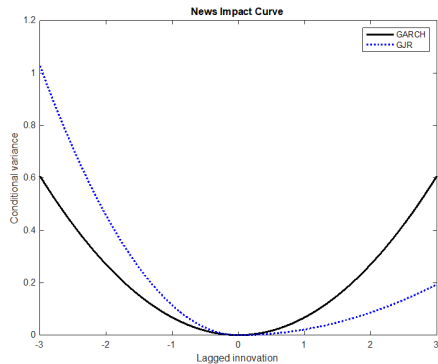
In a symmetric GARCH model:

$$f(+x, \sigma^2) = f(-x, \sigma^2). \quad (40)$$

With leverage effects:

$$f(-x, \sigma^2) > f(+x, \sigma^2), \quad x > 0. \quad (41)$$

News impact curve



The GJR model adds a sign effect:

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \gamma \mathbb{1}_{\{\varepsilon_{t-1} < 0\}} \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2. \quad (42)$$

Interpretation:

- positive shock: coefficient is α ,
- negative shock: coefficient is $\alpha + \gamma$,
- if $\gamma > 0$, negative shocks increase volatility more.

With symmetric innovations, covariance stationarity requires approximately:

$$\alpha + \frac{\gamma}{2} + \beta < 1. \quad (43)$$

The threshold GARCH model is written on volatility rather than variance:

$$\sigma_t = \omega + \alpha|\varepsilon_{t-1}| + \gamma\mathbb{1}_{\{\varepsilon_{t-1} < 0\}}|\varepsilon_{t-1}| + \beta\sigma_{t-1}. \quad (44)$$

The idea is the same:

- size matters through $|\varepsilon_{t-1}|$,
- sign matters through the indicator variable,
- the model is easy to interpret as a response curve.

The EGARCH model works with log variance:

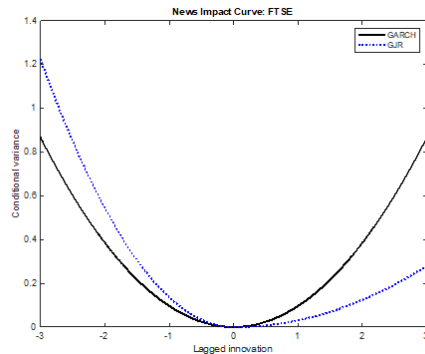
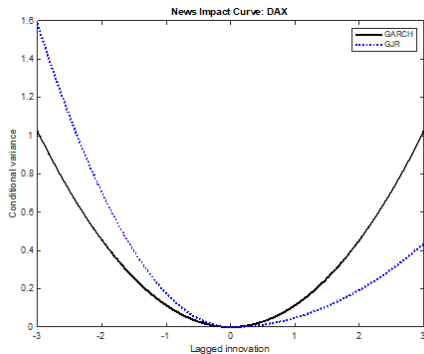
$$\log(\sigma_t^2) = \omega + \alpha (|z_{t-1}| - \mathbb{E}|z_{t-1}|) + \gamma z_{t-1} + \beta \log(\sigma_{t-1}^2). \quad (45)$$

This has two practical advantages:

- σ_t^2 is automatically positive,
- the sign effect is explicit through γz_{t-1} .

If $\gamma < 0$, negative shocks increase future volatility more than positive shocks.

News impact curves across markets



Sometimes volatility affects expected returns directly:

$$r_t = \mu + \lambda \sigma_t^2 + \varepsilon_t, \quad (46)$$

$$\varepsilon_t = \sigma_t Z_t. \quad (47)$$

The parameter λ measures a risk-return trade-off:

- $\lambda > 0$: investors require higher expected returns when risk is high,
- $\lambda = 0$: no direct conditional risk premium,
- $\lambda < 0$: high volatility coincides with lower expected returns.

Empirically, this relation is often unstable at daily frequency.

Non-Gaussian GARCH

Why Gaussian innovations are often insufficient

A Gaussian GARCH model can generate unconditional fat tails because volatility changes through time.

But in many financial series, this is not enough:

- standardized residuals still have fat tails,
- extreme negative returns are too frequent,
- Value-at-Risk violations are clustered,
- density forecasts are too thin-tailed.

The solution is to keep the GARCH variance equation and change the innovation distribution.

The general model is:

$$r_t = \mu_t + \varepsilon_t, \quad (48)$$

$$\varepsilon_t = \sigma_t z_t, \quad (49)$$

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2, \quad (50)$$

$$z_t \sim g(z_t \mid \eta). \quad (51)$$

The distribution g is standardized:

$$\mathbb{E}[z_t] = 0, \quad \text{Var}(z_t) = 1. \quad (52)$$

The parameter vector η controls fat tails and possibly asymmetry.

Conditional versus unconditional moments

If:

$$\varepsilon_t = \sigma_t z_t, \quad (53)$$

then unconditional higher moments combine two effects: the time variation of σ_t and the distribution of z_t .

For kurtosis:

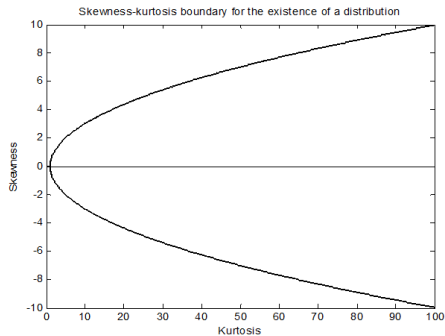
$$K_u = \frac{\mathbb{E}[\varepsilon_t^4]}{\mathbb{E}[\varepsilon_t^2]^2} = \frac{\mathbb{E}[\sigma_t^4]\mathbb{E}[z_t^4]}{\mathbb{E}[\sigma_t^2]^2}. \quad (54)$$

Since:

$$\mathbb{E}[\sigma_t^4] \geq \mathbb{E}[\sigma_t^2]^2, \quad (55)$$

time-varying volatility mechanically increases unconditional kurtosis.

Skewness-kurtosis boundary



Not every pair of skewness and kurtosis can come from a valid distribution.

The most common non-Gaussian GARCH specification is Student-t:

$$z_t \sim t_\nu^*, \quad (56)$$

where t_ν^* is standardized to have variance one.

The standardized density is:

$$g(z \mid \nu) = \frac{\Gamma((\nu + 1)/2)}{\Gamma(\nu/2)\sqrt{\pi(\nu - 2)}} \left(1 + \frac{z^2}{\nu - 2}\right)^{-(\nu+1)/2}. \quad (57)$$

The parameter ν controls tail thickness:

- low ν : fat tails,
- high ν : close to Gaussian.

Fat tails are not the only issue. Equity returns can also be asymmetric.

Two practical choices are:

1. **Skewed Student:** adds one skewness parameter to Student-t innovations.
2. **Mixture of Gaussians:** combines several normal densities:

$$g(z) = \sum_{k=1}^K \pi_k \phi(z; \mu_k, \sigma_k^2), \quad \sum_{k=1}^K \pi_k = 1. \quad (58)$$

These models can fit both tails more flexibly, but they also increase estimation risk.

Generalized hyperbolic distributions

The generalized hyperbolic family is a flexible class for financial returns.

It can generate:

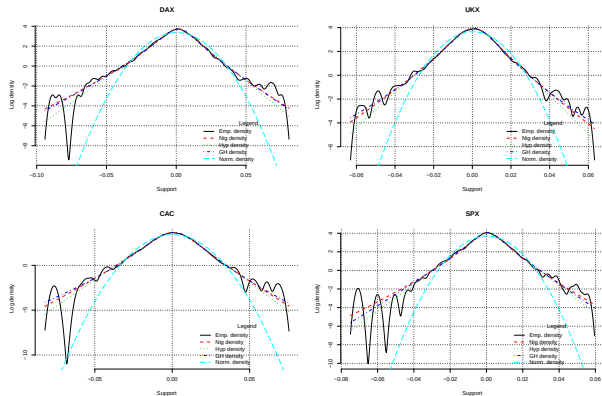
- fat tails,
- skewness,
- semi-heavy tails,
- several well-known special cases.

The practical message is:

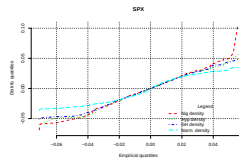
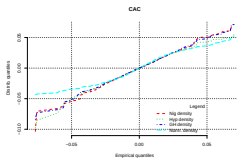
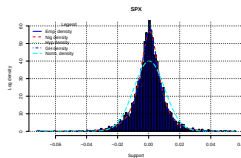
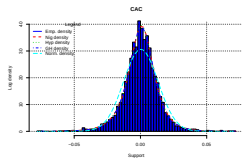
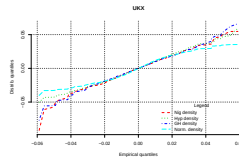
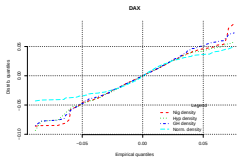
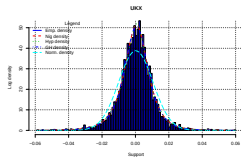
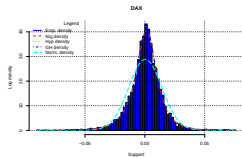
more flexible density $\neq$ better forecast. (59)

A richer conditional density must be judged out-of-sample, not only by in-sample fit.

Non-Gaussian densities in practice



Histogram and Q-Q plots



Likelihood with a generic conditional distribution

For any standardized density g , the log-likelihood contribution is:

$$\log \ell_t(\theta, \eta) = -\frac{1}{2} \log(\sigma_t^2(\theta)) + \log g \left(\frac{r_t - \mu_t(\theta)}{\sigma_t(\theta)} \mid \eta \right). \quad (60)$$

The full log-likelihood is:

$$\log L(\theta, \eta) = \sum_{t=1}^T \log \ell_t(\theta, \eta). \quad (61)$$

The estimation problem now includes both volatility parameters and distribution parameters.

Quasi maximum likelihood

If the Gaussian distribution is wrong, Gaussian ML is interpreted as quasi-ML.

The Gaussian QMLE can remain consistent for the GARCH variance parameters under regularity conditions, but the usual standard errors are no longer reliable.

Use the sandwich covariance matrix:

$$\Omega = A_0^{-1} B_0 A_0^{-1}, \quad (62)$$

where:

$$A_0 = \mathbb{E} \left[-\frac{\partial^2 \log \ell_t(\theta_0)}{\partial \theta \partial \theta'} \right], \quad (63)$$

$$B_0 = \mathbb{E} \left[\frac{\partial \log \ell_t(\theta_0)}{\partial \theta} \frac{\partial \log \ell_t(\theta_0)}{\partial \theta'} \right]. \quad (64)$$

For two density models, compare log-likelihood contributions:

$$d_t = \log \ell_{1,t} - \log \ell_{2,t}. \quad (65)$$

The null hypothesis is:

$$H_0 : \mathbb{E}[d_t] = 0. \quad (66)$$

This is a density comparison, not only a volatility comparison.

A model with a better conditional density can still have a similar variance forecast.

Evaluating volatility forecasts

The Patton problem

Volatility is latent:

$$\sigma_{t+1}^2 \text{ is not observed.} \quad (67)$$

Instead, we evaluate forecasts using a proxy, such as:

$$\tilde{\sigma}_{t+1}^2 = r_{t+1}^2, \quad (68)$$

or a realized variance measure from high-frequency data.

But proxies are noisy:

$$\tilde{\sigma}_{t+1}^2 = \sigma_{t+1}^2 + \text{measurement error.} \quad (69)$$

The forecast ranking can depend on the loss function.

Patton's key lesson is that not every loss function is safe when the volatility proxy is noisy.

Two commonly used robust losses for variance forecasts are:

$$L_{\text{MSE}}(\tilde{\sigma}^2, \hat{\sigma}^2) = (\tilde{\sigma}^2 - \hat{\sigma}^2)^2, \quad (70)$$

$$L_{\text{QLIKE}}(\tilde{\sigma}^2, \hat{\sigma}^2) = \log(\hat{\sigma}^2) + \frac{\tilde{\sigma}^2}{\hat{\sigma}^2}. \quad (71)$$

In practice, QLIKE is often preferred because it penalizes too-low variance forecasts strongly, which is useful in risk management.

Comparing two volatility forecasts

Let $\hat{\sigma}_{1,t+1}^2$ and $\hat{\sigma}_{2,t+1}^2$ be two forecasts.

Define the loss differential:

$$d_{t+1} = L(\tilde{\sigma}_{t+1}^2, \hat{\sigma}_{1,t+1}^2) - L(\tilde{\sigma}_{t+1}^2, \hat{\sigma}_{2,t+1}^2). \quad (72)$$

The null hypothesis is:

$$H_0 : \mathbb{E}[d_{t+1}] = 0. \quad (73)$$

A simple Diebold-Mariano statistic is:

$$DM = \frac{\bar{d}}{\widehat{\text{se}}(\bar{d})}. \quad (74)$$

For an empirical volatility comparison:

1. decide what is being forecast: variance, volatility, VaR or full density,
2. use an out-of-sample design,
3. use at least one robust loss function,
4. report economic interpretation, not only statistical significance,
5. check whether the result is dominated by a crisis episode.

A good classroom benchmark is:

Gaussian GARCH versus Student-t GARCH. (75)

Empirical application: S&P 500

Application: objective

We use the S&P 500 price series provided with the course.

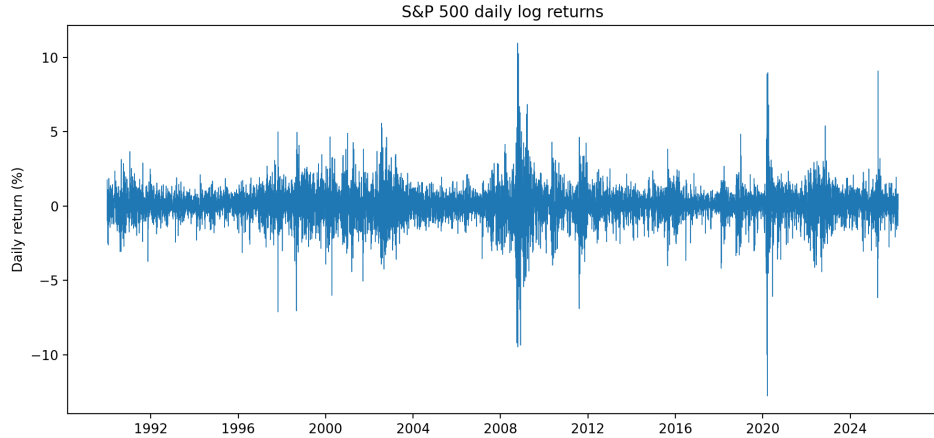
The application illustrates four points:

1. how to compute daily log returns,
2. how to estimate a Gaussian GARCH(1,1),
3. how to estimate a Student-t GARCH(1,1),
4. how to compare volatility forecasts with MSE and QLIKE.

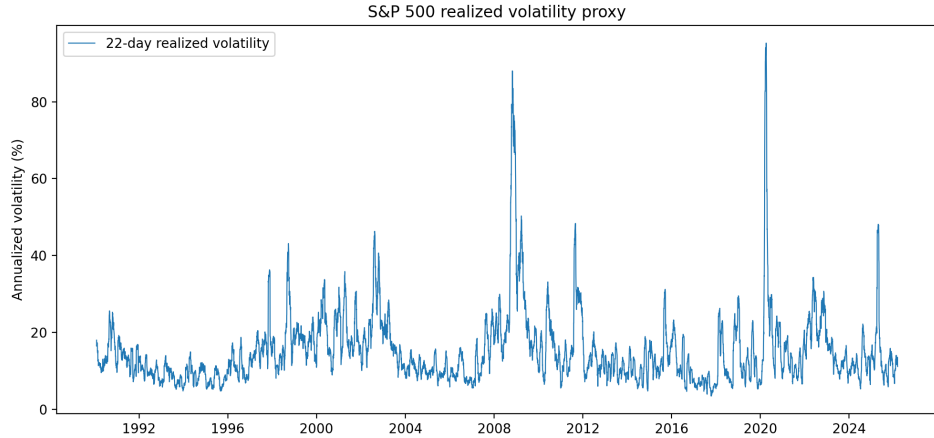
The code is intentionally simple:

no black-box GARCH package is required. (76)

Application data



Realized volatility proxy



For both models:

$$r_t = \mu + \varepsilon_t, \quad (77)$$

$$\varepsilon_t = \sigma_t z_t, \quad (78)$$

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2. \quad (79)$$

Two assumptions are compared:

$$z_t \sim N(0, 1), \quad (80)$$

$$z_t \sim t_\nu^*. \quad (81)$$

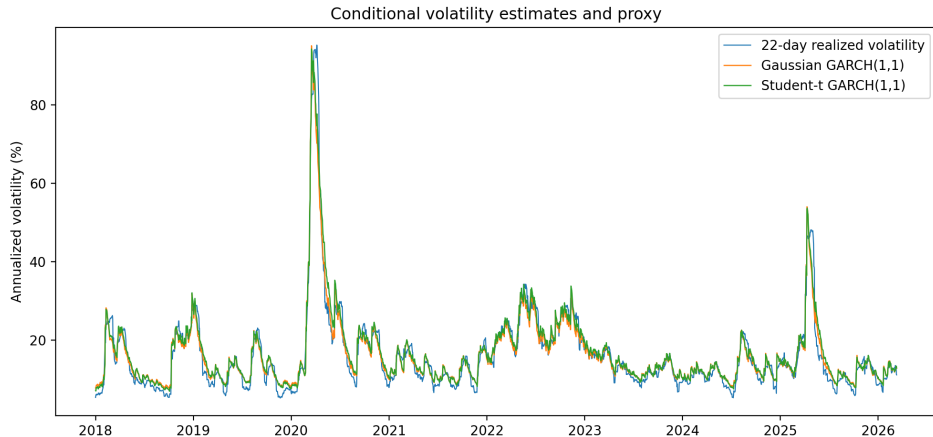
The training sample ends in 2018. Forecasts are evaluated from 2019 to 2026.

Model	$\hat{\omega}$	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\alpha} + \hat{\beta}$	ν	AIC
Gaussian	0.0126	0.083	0.906	0.989	–	19623
Student-t	0.0065	0.079	0.920	0.999	5.4	19176

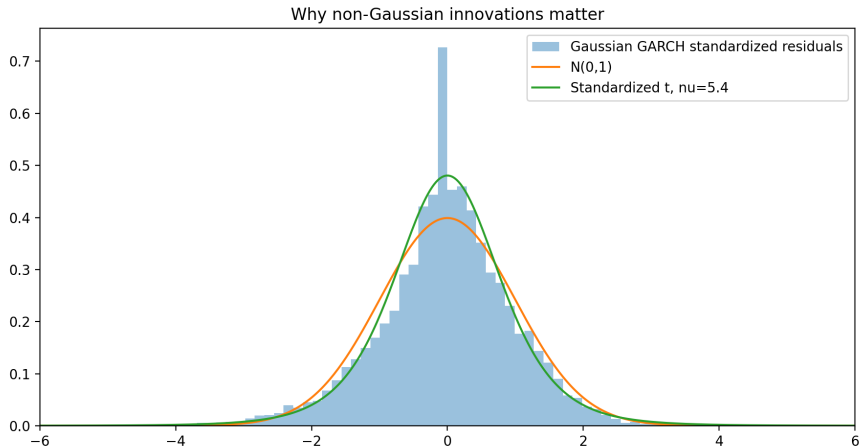
Training sample: 1990-01-02 to 2018-12-31. Evaluation sample: 2019-01-01 to 2026-03-11.

The Student-t model improves the in-sample density fit, mainly by allowing fatter tails.

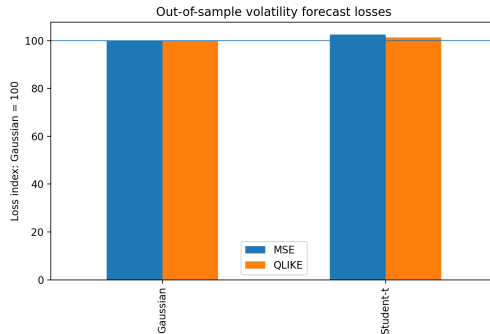
Conditional volatility estimates



Standardized residuals and fat tails



Out-of-sample volatility forecast losses



Average QLIKE: Gaussian = 0.916, Student-t = 0.928. Average MSE: Gaussian = 32.13, Student-t = 32.96.

How the notebook maps to the lecture

The notebook follows the same logic as the course:

1. build the return series,
2. write the GARCH recursion,
3. write the Gaussian and Student-t log-likelihoods,
4. estimate parameters by numerical optimization,
5. compute one-step-ahead variance forecasts,
6. evaluate forecasts with Patton-style losses.

The central lesson is:

better density fit $\neq$ automatically better volatility forecast. (82)

Key takeaways

1. GARCH models are parsimonious models of volatility persistence.
2. Gaussian GARCH is a useful baseline, but standardized residuals are often fat-tailed.
3. Non-Gaussian GARCH models improve conditional density modeling.
4. Asymmetric GARCH models capture the stronger volatility response to bad news.
5. Volatility forecasts should be evaluated out-of-sample with robust loss functions.
6. For students: the recursion is more important than the software package.